

PROJECT REPORT

Vulnerability Assessment and Penetration Testing (VAPT) on a Web Application

Internship Project Report

Project Domain: Cybersecurity

Testing Environment: Kali Linux

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Submitted By

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Disclaimer: This report has been created strictly for educational purposes under the IBM–NASSCOM PBEL Cyber Security Program. All vulnerability testing was performed on a locally hosted vulnerable web application in a controlled environment. No live systems or production servers were tested. The goal of this project is to learn ethical hacking, secure coding practices, and web security analysis techniques.

Abstract: This project, under the IBM–NASSCOM Project-Based Experiential Learning (PBEL) Program, focuses on performing Vulnerability Assessment and Penetration Testing (VAPT) on a vulnerable web application called DVWA (Damn Vulnerable Web Application). The project involves identifying and exploiting real-world vulnerabilities based on the OWASP Top 10, including SQL Injection, XSS, Command Injection, CSRF, and more. Testing was performed locally using XAMPP, and tools like Burp Suite Community Edition and its embedded Chromium browser. Each vulnerability was tested, documented with payloads and screenshots, and appropriate mitigation strategies were provided. The project aims to build practical cyber security skills and demonstrate the importance of secure web application development.

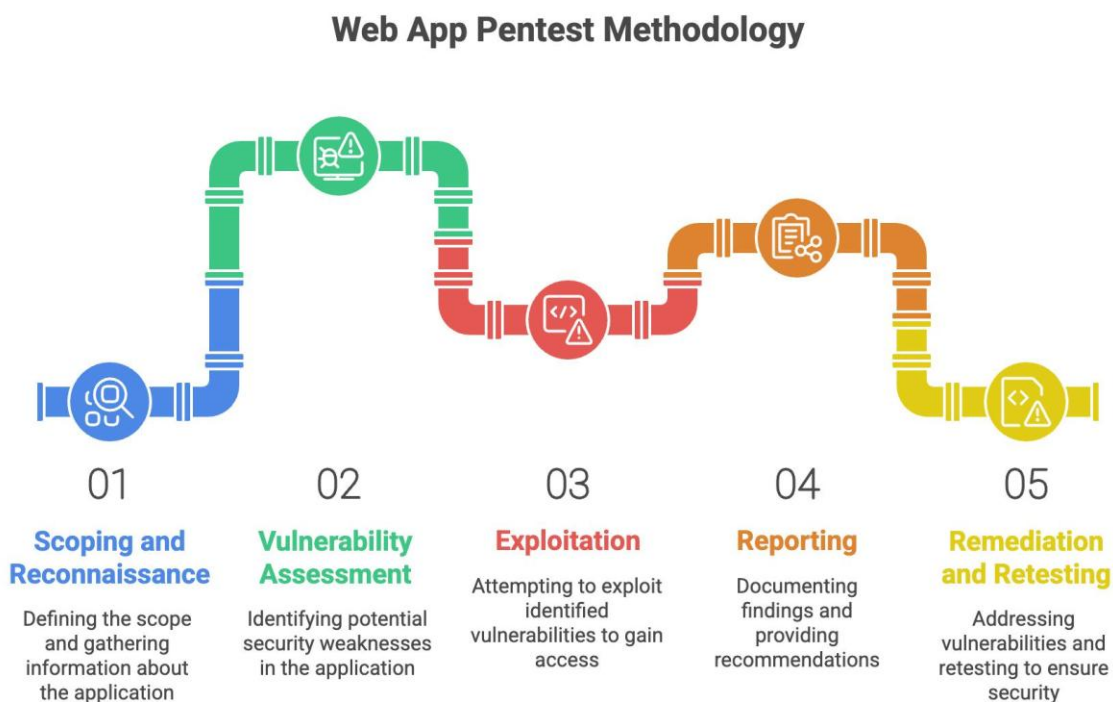
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VULNERABILITY ASSESSMENT AND PENETRATION TESTING (VAPT) ON A WEB APPLICATION

1. Introduction

Web applications have become an essential part of modern organizations for providing online services such as e-commerce, banking, education, healthcare, and social networking. Since these applications store sensitive user information, they are common targets for cyberattacks. Vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), Broken Authentication, and Insecure Direct Object References (IDOR) can allow attackers to steal data, modify application functionality, or gain unauthorized access.

Vulnerability Assessment and Penetration Testing (VAPT) is a comprehensive security assessment methodology used to identify, verify, and exploit security weaknesses in a web application under authorized conditions. Vulnerability Assessment identifies potential security flaws, while Penetration Testing validates whether those vulnerabilities can be successfully exploited by an attacker.



2. Objectives

The objectives of this project are:

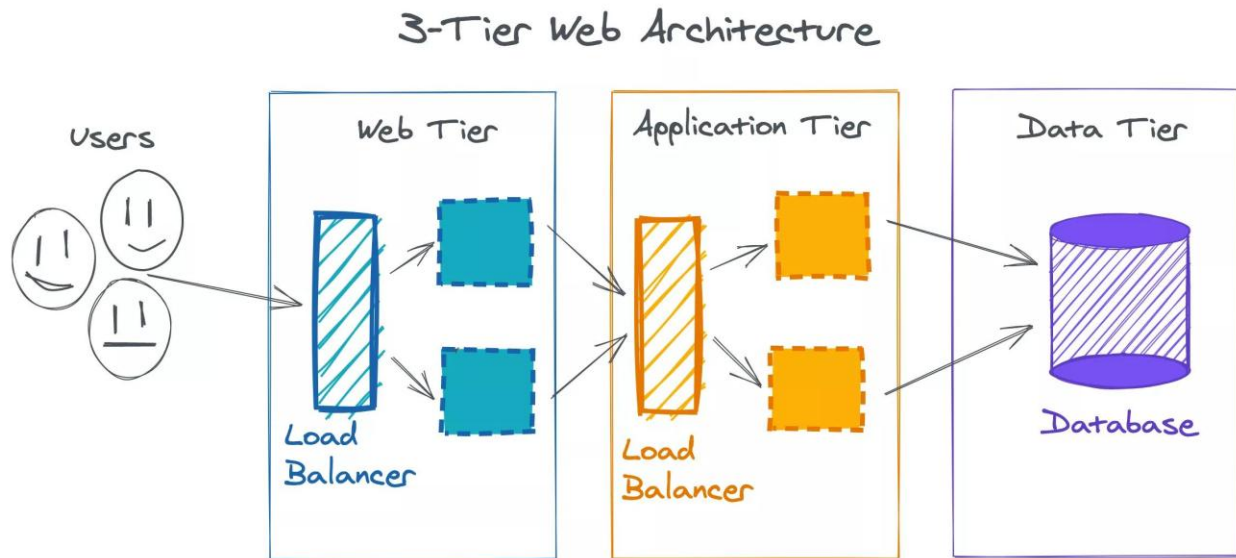
- Identify security vulnerabilities in the web application.
 - Analyze the impact of discovered vulnerabilities.
 - Demonstrate the exploitation of vulnerabilities in a controlled environment.
 - Provide recommendations to mitigate identified security risks.
 - Improve the overall security of the application.
-

3. Scope

The assessment includes the following components:

- Login and Authentication System
- User Registration
- User Dashboard
- Search Functionality
- File Upload Module
- Contact Form
- Session Management
- Database Interaction
- HTTP Headers
- Server Configuration

The assessment excludes denial-of-service attacks and testing of third-party services not owned by the organization.

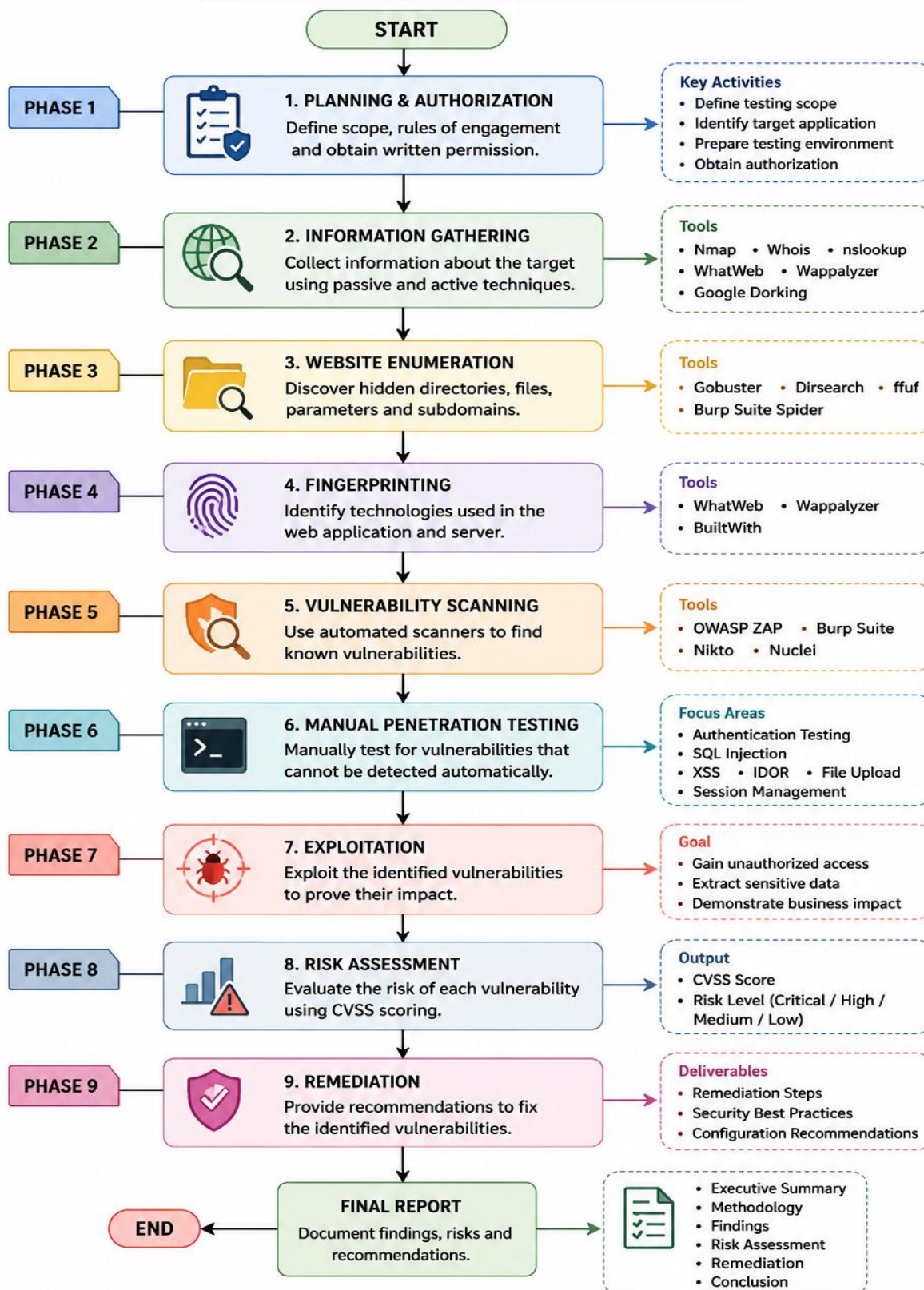


4. Methodology

The VAPT process follows the following phases:

1. Planning and Authorization
2. Information Gathering
3. Reconnaissance
4. Vulnerability Scanning
5. Manual Testing
6. Exploitation
7. Risk Analysis
8. Reporting
9. Remediation

VAPT PROCESS FLOWCHART



Phase 1: Planning and Authorization

Before starting the assessment, proper authorization must be obtained from the application owner. Testing without permission is illegal and unethical.

Activities Performed

- Define testing scope.
- Identify target URL/IP.
- Prepare testing environment.
- Install required tools.
- Obtain written permission.

Deliverables

- Scope Document
 - Authorization Letter
 - Testing Plan
-

Phase 2: Information Gathering (Reconnaissance)

Information gathering is the first technical phase of VAPT. The purpose is to collect as much information as possible about the target application.

Information Collected

- Target IP Address
- Domain Name
- DNS Records
- Server Technology
- Programming Language
- Framework

- Database
- Open Ports
- Running Services

Tools Used

- Nmap
- Whois
- nslookup
- WhatWeb
- Wappalyzer

Commands

nmap -A target-ip

Aim: An aggressive scan (-A) collects detailed information by enabling OS detection, service/version detection, scripts, and traceroute in a single command.

```
(kali㉿kali)-[~]
$ nmap -A 104.21.77.200
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-14 01:01 EST
Nmap scan report for 104.21.77.200
Host is up (0.0048s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
80/tcp    open  http           Cloudflare http proxy
|_http-title: Site doesn't have a title (text/plain; charset=UTF-8).
|_http-server-header: cloudflare
443/tcp   open  ssl/https      cloudflare
|_http-title: 400 The plain HTTP request was sent to HTTPS port
|_http-server-header: cloudflare
8080/tcp   open  http           Cloudflare http proxy
|_http-title: Site doesn't have a title (text/plain; charset=UTF-8).
|_http-server-header: cloudflare
8443/tcp   open  ssl/https-alt?
|_http-title: 400 The plain HTTP request was sent to HTTPS port
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (88%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (88%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1   0.70 ms  104.21.77.200

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 35.70 seconds
```

whatweb <http://target-url>

```
(kali@kali)-[~]
$ whatweb uptoskills.com
http://uptoskills.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[cloudflare], IP[104.21.77.200], RedirectLocation[https://uptoskills.com/
UncommonHeaders[speculation-rules,report-to,nel,cf-ray,alt-svc]
https://uptoskills.com [200 OK] Country[UNITED STATES][US], Email[info@uptoskills.com], HTML5, HTTPServer[cloudflare], IP[104.21.77.200], Meta-Author[Uptoski
2090,IN,India,New Delhi], Open-Graph-Protocol[website], Script[application/ld+json,module], Strict-Transport-Security[max-age=31536000; includeSubDomains; pr
s AI-Powered Hiring & Career Ecosystem], UncommonHeaders[report-to,nel,cf-cache-status,access-control-allow-origin,x-permitted-cross-domain-policies,link,con
n-opener-policy,cross-origin-resource-policy,permissions-policy,referrer-policy,x-content-type-options,x-download-options,speculation-rules,cf-ray,alt-svc],
tection[1; mode=block]
https://uptoskills.com/ [200 OK] Country[RESERVED][ZZ], Email[info@uptoskills.com], HTML5, HTTPServer[cloudflare], IP[172.67.211.113], Meta-Author[Uptoskills
0,IN,India,New Delhi], Open-Graph-Protocol[website], Script[application/ld+json,module], Strict-Transport-Security[max-age=31536000; includeSubDomains; prelo
I-Powered Hiring & Career Ecosystem], UncommonHeaders[report-to,nel,cf-cache-status,access-control-allow-origin,x-permitted-cross-domain-policies,link,conten
pener-policy,cross-origin-resource-policy,permissions-policy,referrer-policy,x-content-type-options,x-download-options,speculation-rules,cf-ray,alt-svc], X-F
tion[1; mode=block]
```

whois target-domain

```
(kali@kali)-[~]
$ whois uptoskills.com
Domain Name: UPTOSKILLS.COM
Registry Domain ID: 2549370386_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2026-07-09T11:31:50Z
Creation Date: 2020-07-29T06:26:04Z
Registry Expiry Date: 2027-07-29T06:26:04Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: LENNON.NS.CLOUDFLARE.COM
Name Server: REZA.NS.CLOUDFLARE.COM
DNSSEC: signedDelegation
DNSSEC DS Data: 2371 13 2 E4E18722E69FF26E33CEB5E4F601A9F5DA1F10A814C029081297470FCD5BE7D6
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-17T10:46:17Z <<<
```

For more information on Whois status codes, please visit <https://icann.org/epp>

NOTICE: The expiration date displayed in this record is the date the registrar's sponsorship of the domain name registration in the registry is currently set to expire. This date does not necessarily reflect the expiration date of the domain name registrant's agreement with the sponsoring registrar. Users may consult the sponsoring registrar's Whois database to view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois

Expected Output

- Apache Server
- PHP 8.x
- MySQL Database

- Port 80 Open
 - Port 443 Open
-

Phase 3: Website Enumeration

Enumeration identifies hidden resources inside the application.

Objectives

- Discover hidden directories.
- Find backup files.
- Identify admin panels.
- Detect exposed configuration files.

Tools

- Gobuster
- Dirsearch
- ffuf

Example Command

```
gobuster dir -u http://target.com -w  
/usr/share/wordlists/dirb/common.txt
```

Aim: Find hidden directories and files on a website.

```
(kali@kali)-[~]
$ gobuster dir -u http://uptoskills.com -w /usr/share/wordlists/dirb/common.txt

Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://uptoskills.com
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8.2
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

Progress: 0 / 1 (0.00%)
2026/03/06 04:07:54 the server returns a status code that matches the provided options for non existing urls. http://uptoskills.com/23584606-75d7-48e6-b6f4-341af988201a => 301 (redirect to https://uptoskills.com/23584606-75d7-48e6-b6f4-341af988201a) (Length: 0). Please exclude the response length or the status code or set the wildcard option.. To continue please exclude the status code or the length
```

Findings

The scan may reveal:

- /admin
- /login
- /uploads
- /backup
- /config

Phase 4: Fingerprinting

Fingerprinting identifies technologies used to build the application.

Information Identified

- Web Server
- Programming Language
- CMS
- Framework
- Database
- Operating System

```
(kali㉿kali)~]
$ whatweb uptoskills.com
http://uptoskills.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[cldflare], IP[104.21.77.200], RedirectLocation[https://uptoskills.com/UncommonHeaders[speculation-rules,report-to,nel,cf-ray,alt-svc]
https://uptoskills.com [200 OK] Country[UNITED STATES][US], Email[info@uptoskills.com], HTML5, HTTPServer[cldflare], IP[104.21.77.200], Meta-Author[Uptoski 2090,IN,India,New Delhi], Open-Graph-Protocol[website], Script[application/ld+json,module], Strict-Transport-Security[max-age=31536000; includeSubDomains; pr s AI-Powered Hiring & Career Ecosystem], UncommonHeaders[report-to,nel,cf-cache-status,access-control-allow-origin,x-permitted-cross-domain-policies,link,con n-opener-policy,cross-origin-resource-policy,permissions-policy,referrer-policy,x-content-type-options,x-download-options,speculation-rules,cf-ray,alt-svc], taction[1; mode=block]
https://uptoskills.com/ [200 OK] Country[RESERVED][22], Email[info@uptoskills.com], HTML5, HTTPServer[cldflare], IP[172.67.211.113], Meta-Author[Uptoskills 0,TN,India,New Delhi], Open-Graph-Protocol[website], Script[application/ld+json,module], Strict-Transport-Security[max-age=31536000; includeSubDomains; prelo I-Powered Hiring & Career Ecosystem], UncommonHeaders[report-to,nel,cf-cache-status,access-control-allow-origin,x-permitted-cross-domain-policies,link,conten pener-policy,cross-origin-resource-policy,permissions-policy,referrer-policy,x-content-type-options,x-download-options,speculation-rules,cf-ray,alt-svc], X-F tion[1; mode=block]
```

The screenshot shows a web browser window with a search bar containing 'uptoskills.com'. The search results page displays 'uptoskills.com' as the top result. A Wappalizer overlay is visible on the right side of the browser window, displaying various website analytics and management tools. The overlay includes sections for 'Analytics' (Google Analytics), 'Tag managers' (Google Tag Manager), 'Security' (HSTS), 'Reverse proxies' (Envoy), 'Web servers' (Apache Traffic Server), and a section for 'Automate technology lookups'.

 **Wappalyzer**

TECHNOLOGIES

MORE INFO

Export

Analytics
 [Google Analytics](#) GA4

Tag managers
 [Google Tag Manager](#)

Security
 [HSTS](#)

Reverse proxies
 [Envoy](#)

Web servers
 [Apache Traffic Server](#)

[Something wrong or missing?](#)

Generate sales leads

Find new prospects by the technologies they use. Reach out to customers of Shopify, Magento, Salesforce and others.

• BuiltWith

uptoskills.com

Search for opportunities, profiles, leagues...

Transform Your Talent Acquisition!

Let's Make Freshers Employable!

Connecting Colleges, Candidates, and Corporates through an Innovative Tech Platform

Start My Journey

Contact Us

UPTOSKILLS PRO

Your All-in-One

Your privacy is important to us

Explore Pro

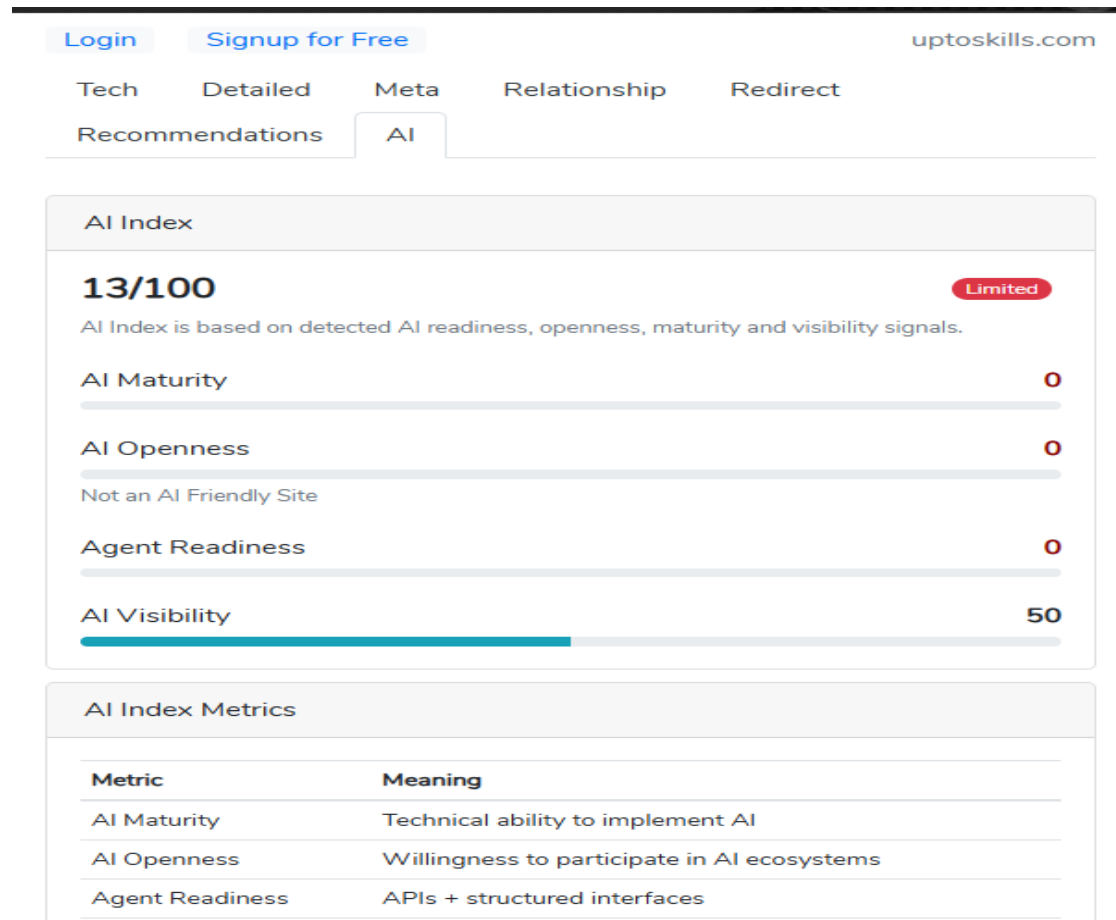
UPTOSKILLS.COM

Analytics and Tracking

 Google Analytics Audience Measurement	Sep 2020	Jun 2026
 Google Analytics 4	Nov 2025	Jun 2026
 Google Universal Analytics	Sep 2020	Mar 2022
 Global Site Tag Tag Management Conversion Tracking Advertiser Tracking	Apr 2021	Jul 2026
 Cloudflare Insights Application Performance: Audience Measurement Visitor Count Tracking Web Analytics +6 more	Nov 2025	Jul 2026
 Cloudflare Web Analytics Audience Measurement Visitor Count Tracking	Nov 2025	Jul 2026

Widgets

 Periodic Schedule Management Appointment Booking Call Scheduling	Sep 2025	Apr 2026
 Google Tag Manager Tag Management Marketing Automation Data Management Platform	Apr 2021	Jun 2026

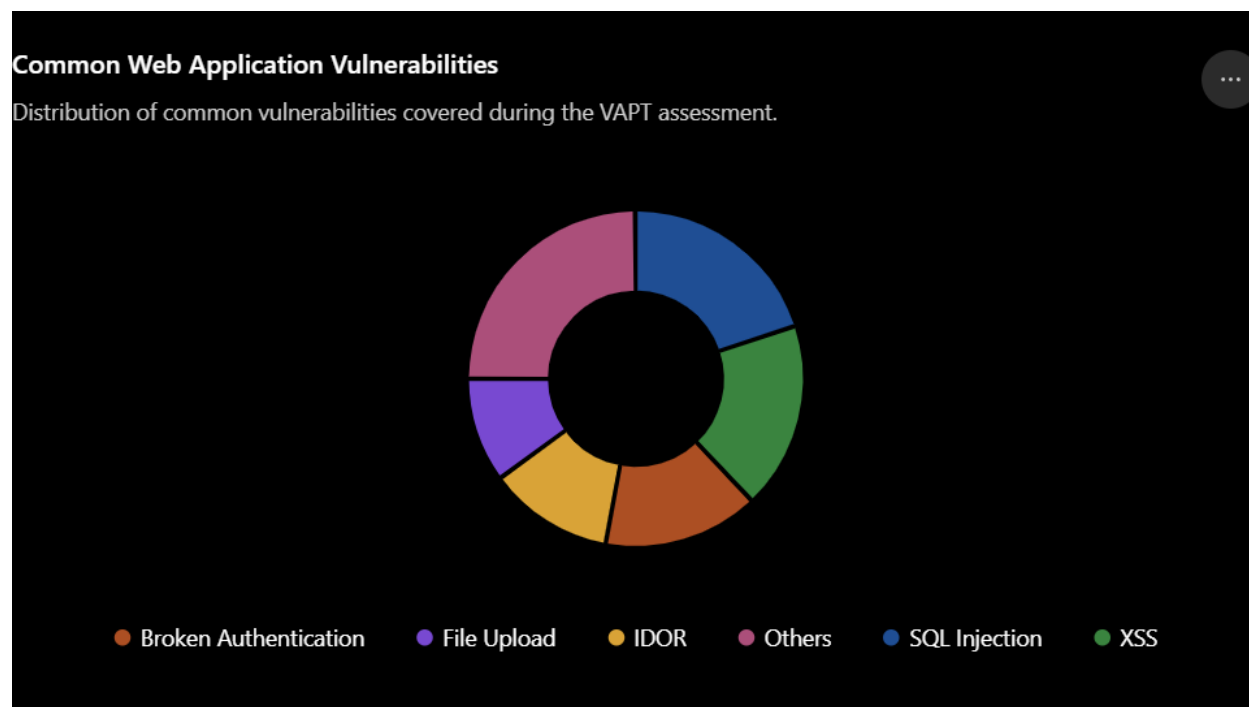


Example Output

- Apache 2.4
- PHP
- Bootstrap
- MySQL
- Linux Ubuntu

Phase 5: Vulnerability Assessment

Automated scanners identify known vulnerabilities.



Tools Used

- OWASP ZAP
- Burp Suite
- Nikto
- Nuclei

Nikto Scan

- **nikto -h <http://target.com>**


```
(kali@kali)-[~]
$ nikto -h https://uptoskills.com
- Nikto v2.5.0

+ Multiple IPs found: 104.21.77.200, 172.67.211.113, 2606:4700:3033::6815:4dc8, 2606:4700:3035::ac43:d371
+ Target IP: 104.21.77.200
+ Target Hostname: uptoskills.com
+ Target Port: 443

+ SSL Info: Subject: /CN=uptoskills.com
           Ciphers: TLS_AES_256_GCM_SHA384
           Issuer: /C=US/O=Google Trust Services/CN=WE1
+ Start Time: 2026-02-28 06:15:43 (GMT-5)

+ Server: cloudflare
+ /: Retrieved access-control-allow-origin header: *.
+ /: Uncommon header 'server-timing' found, with multiple values: (cfCacheStatus;desc="DYNAMIC",cfEdge;dur=6,cfOrigin;du
+ /: Uncommon header 'speculation-rules' found, with contents: "/cdn-cgi/speculation".
+ /: An alt-svc header was found which is advertising HTTP/3. The endpoint is: ':443'. Nikto cannot test HTTP/3 over QUI
lla.org/en-US/docs/Web/HTTP/Headers/alt-svc
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ /robots.txt: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of th
to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ /robots.txt: contains 23 entries which should be manually viewed. See: https://developer.mozilla.org/en-US/docs/Glossa
+ /: The Content-Encoding header is set to "deflate" which may mean that the server is vulnerable to the BREACH attack.
+ ERROR: Error limit (20) reached for host, giving up. Last error: opening stream: can't connect: SSL negotiation failed
:ssl/tls alert handshake failure at /var/lib/nikto/plugins/LW2.pm line 5254.
; at /var/lib/nikto/plugins/LW2.pm line 5254.
+ Scan terminated: 20 error(s) and 7 item(s) reported on remote host
+ End Time: 2026-02-28 06:17:28 (GMT-5) (105 seconds)

+ 1 host(s) tested
```

Vulnerabilities Detected

- Missing Security Headers
- Server Version Disclosure
- Directory Listing Enabled
- Outdated Software
- Weak SSL Configuration

Phase 6: Manual Penetration Testing

Automated scanners cannot identify every vulnerability. Therefore, manual testing is performed.

6.1 Authentication Testing

Objective

Evaluate authentication mechanisms.

Tests Performed

- Weak Password Testing
- Default Credentials
- Password Complexity
- Password Reset
- Account Lockout
- Multi-factor Authentication

Expected Result

Only authorized users should successfully log in.

6.2 SQL Injection Testing

Objective

Determine whether user inputs interact securely with the database.

Tools

- SQLMap
- Burp Suite

Example SQL Injection Payload

' OR '1'='1

SQLMap Command

```
• sqlmap -u "http://target.com/product.php?id=1" -dbs
```

Project: VAPT on a Web Application

```
(kali@kali)~$  
$ sqlmap -u "http://uptoskills.com/page?id=1" --dbs  
 {1.10.2#stable}  
https://sqlmap.org  
  
[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume n  
responsible for any misuse or damage caused by this program  
  
[*] starting @ 10:31:40 /2026-03-13/  
  
[10:31:40] [INFO] testing connection to the target URL  
[10:31:40] [WARNING] potential permission problems detected ('Access denied')  
[10:31:40] [WARNING] the web server responded with an HTTP error code (403) which could interfere with the results of the tests  
[10:31:40] [INFO] testing if the target URL content is stable  
[10:31:41] [WARNING] target URL content is not stable (i.e. content differs). sqlmap will base the page comparison on a sequence matcher. If no dynamic nor injectable parameters are detected, or in case of junk  
manual paragraph 'Page comparison'  
how do you want to proceed? [(C)ontinue/(s)tring/(r)egex/(q)uit] c  
[10:31:45] [INFO] testing if GET parameter 'id' is dynamic  
[10:31:45] [WARNING] GET parameter 'id' does not appear to be dynamic  
[10:31:45] [WARNING] heuristic (basic) test shows that GET parameter 'id' might not be injectable  
[10:31:45] [INFO] testing for SQL injection on GET parameter 'id'  
[10:31:45] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause'  
[10:31:46] [INFO] testing 'Boolean-based blind - Parameter replace (original value)'  
[10:31:46] [INFO] testing 'MySQL >= 5.1 AND error-based - WHERE, HAVING, ORDER BY or GROUP BY clause (EXTRACTVALUE)'  
[10:31:46] [INFO] testing 'PostgreSQL AND error-based - WHERE or HAVING clause'  
[10:31:46] [INFO] testing 'Microsoft SQL Server/Sybase AND error-based - WHERE or HAVING clause (IN)'  
[10:31:46] [INFO] testing 'Oracle AND error-based - WHERE or HAVING clause (XMLType)'  
[10:31:47] [INFO] testing 'Generic inline queries'  
[10:31:47] [INFO] testing 'PostgreSQL > 8.1 stacked queries (comment)'  
[10:31:47] [INFO] testing 'Microsoft SQL Server/Sybase stacked queries (comment)'  
[10:31:48] [INFO] testing 'Oracle stacked queries (DBMS_PIPE.RECEIVE_MESSAGE - comment)'  
[10:31:48] [INFO] testing 'MySQL >= 5.0.12 AND time-based blind (query SLEEP)'  
[10:31:48] [INFO] testing 'PostgreSQL > 8.1 AND time-based blind'  
[10:31:48] [INFO] testing 'Microsoft SQL Server/Sybase time-based blind (IF)'  
[10:31:49] [INFO] testing 'Oracle AND time-based blind'  
it is recommended to perform only basic UNION tests if there is not at least one other (potential) technique found. Do you want to reduce the number of requests? [Y/n] y  
[10:31:58] [INFO] testing 'Generic UNION query (NULL) - 1 to 10 columns'  
[10:31:59] [WARNING] GET parameter 'id' does not seem to be injectable
```

Risk

Critical

Impact

- Database Dump
- Credential Theft
- Data Manipulation
- Complete Database Access

Recommendation

- Use Prepared Statements
- Parameterized Queries
- Input Validation

6.3 SSL/TLS Testing

Tools

- SSL Labs

- testssl.sh

Tests

- HTTPS Enabled
- TLS Version
- Weak Cipher Suites
- Certificate Validity

Phase 7: Exploitation

After identifying vulnerabilities, controlled exploitation is performed to verify their existence.

Examples

- Successfully retrieve database names using SQL Injection.
- Execute JavaScript alert using XSS.
- Access another user's profile through IDOR.
- Upload a test file to demonstrate insecure file upload.

Evidence such as screenshots and HTTP requests/responses should be collected during this phase.

Phase 8: Risk Assessment

Each vulnerability is assigned a severity based on the Common Vulnerability Scoring System (CVSS).

Vulnerability	Severity (CVSS Score)	Risk Level
 SQL Injection	9.8	Critical
 Stored XSS	8.8	High
 Broken Authentication	8.1	High
 IDOR	7.5	High
 File Upload	8.6	High
 Missing Security Headers	4.3	Medium
 Directory Listing	3.7	Low

Phase 9: Remediation

SQL Injection

- Use Prepared Statements.
- Implement Parameterized Queries.
- Validate all user inputs.

Cross-Site Scripting

- Encode all output.
- Sanitize user input.
- Deploy a Content Security Policy.

Broken Authentication

- Enforce strong passwords.
- Enable Multi-Factor Authentication.
- Secure session management.

File Upload

- Restrict allowed file types.
- Verify MIME types.
- Scan uploaded files with antivirus.
- Store uploads outside the web root.

Security Headers

Configure the following headers:

- Content-Security-Policy
- Strict-Transport-Security
- X-Frame-Options
- X-Content-Type-Options
- Referrer-Policy

Tools Used

Tool	Purpose
 Nmap	Network Scanning
 WhatWeb	Technology Detection
 Gobuster	Directory Enumeration
 Burp Suite	Intercepting and Manual Testing
 OWASP ZAP	Automated Web Vulnerability Scanning
 Nikto	Web Server Vulnerability Scanning
 SQLMap	SQL Injection Testing
 Nuclei	Template-Based Vulnerability Scanning
 Wireshark	Network Packet Analysis
 Firefox + FoxyProxy	Proxy Configuration

Conclusion:

The VAPT assessment successfully identified multiple security vulnerabilities affecting the confidentiality, integrity, and availability of the web application. Automated scanning combined with manual penetration testing provided a comprehensive evaluation of the application's security posture. Critical issues such as SQL Injection, Cross-Site Scripting (XSS), Broken Authentication, and Insecure Direct Object References (IDOR) demonstrated the importance of secure coding practices, proper input validation, robust authentication mechanisms, and secure server configuration.